Limiting Reactants and Percent Yield

Answer key with worked solutions

A. Understanding

1. Definitions.

> (a) The reactant used up first, which limits the amount of product. (b) A reactant left over when the reaction stops. (c) The maximum amount of product, calculated from the limiting reactant.

2. Why yields are below 100%.

> Any two of: the reaction does not go to completion; side reactions form other products; product is lost during separation or purification.

B. Working in moles

3. 3.0 mol H_2 + 2.0 mol O_2.

> 3.0 mol H_2 × (1 mol O_2 / 2 mol H_2) = 1.5 mol O_2 needed, so (a) **H_2 is limiting**. (b) H_2O = 3.0 mol H_2 × (2 mol H_2O / 2 mol H_2) = **3.0 mol**. (c) O_2 left = 2.0 mol − 1.5 mol = **0.5 mol**.

4. 1.5 mol N_2 + 3.0 mol H_2.

> 1.5 mol N_2 × (3 mol H_2 / 1 mol N_2) = 4.5 mol H_2 needed, but only 3.0 mol is available, so (a) **H_2 is limiting**. (b) NH_3 = 3.0 mol H_2 × (2 mol NH_3 / 3 mol H_2) = **2.0 mol**. (c) N_2 used = 3.0 mol H_2 × (1 mol N_2 / 3 mol H_2) = 1.0 mol, so N_2 left = 1.5 mol − 1.0 mol = **0.5 mol**.

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C. Working in grams

5. 5.40 g Al + 15.0 g Cl_2.

> n(Al) = 5.40 g ÷ 26.98 g/mol = 0.2001 mol; n(Cl_2) = 15.0 g ÷ 70.90 g/mol = 0.2116 mol.
>
> The Al would need 0.2001 mol Al × (3 mol Cl_2 / 2 mol Al) = 0.3002 mol Cl_2, but only 0.2116 mol is available, so (a) **Cl_2 is limiting**.
>
> (b) n($AlCl_3$) = 0.2116 mol Cl_2 × (2 mol $AlCl_3$ / 3 mol Cl_2) = 0.1410 mol; M($AlCl_3$) = 133.33 g/mol; mass = 0.1410 mol × 133.33 g/mol = **18.8 g**.
>
> (c) Al used = 0.1410 mol; Al left = 0.2001 mol − 0.1410 mol = 0.0591 mol; 0.0591 mol × 26.98 g/mol = **1.59 g**.

6. Percent yield (15.2 g collected).

> Percent yield = 15.2 g ÷ 18.81 g × 100% = **80.8%** (using the unrounded theoretical yield, 18.81 g; the units of g cancel).

7. 10.0 g CH_4 + 32.0 g O_2.

> n(CH_4) = 10.0 g ÷ 16.04 g/mol = 0.6234 mol; n(O_2) = 32.0 g ÷ 32.00 g/mol = 1.000 mol. The methane needs 0.6234 mol CH_4 × (2 mol O_2 / 1 mol CH_4) = 1.247 mol O_2, so (a) **O_2 is limiting**.
>
> (b) n(CO_2) = 1.000 mol O_2 × (1 mol CO_2 / 2 mol O_2) = 0.500 mol; mass = 0.500 mol × 44.01 g/mol = **22.0 g**.
>
> (c) CH_4 used = 0.500 mol × 16.04 g/mol = 8.02 g; CH_4 left = 10.0 g − 8.02 g = **1.98 g**.

D. Challenge

8. A 108% yield.

> Most likely the product was still **wet** or contained **impurities**, which added mass. Mass cannot be created, so a pure, dry product can never exceed the theoretical yield.

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